

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Substitution: $W = \frac{1}{2} \times 0.15 \times 12$ (1) $= 0.9$ [J] (1)	1	1		2	2	
		(ii)		Substitution: 0.9 (ecf) $= 0.5 \times 0.075 \times v^2$ (1) Rearrangement: $v^2 = 24$ (1) $v = 4.9$ (1) m/s (1)	1 1	1 1		4	3	
		(iii)		Because some energy is lost (1) To {heat / friction / resistive forces} (1)	1	1		2		
	(b)			Substitution: $0.8 = F \times 1.5$ (1) $F = \frac{0.8}{1.5}$ (1) $F = 0.5[3]$ [N] (1)	1	1 1		3	3	
	(c)			[For the same extension] twice the force will be required (1) So $\frac{1}{2} F_x$ will be double [so agree] (1)			2	2		
				Question 6 total	5	6	2	13	8	0